SPECIALIZATION MAPS COMMUTE WITH BASE CHANGE OF DISCRETE VALUATION RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Extension of discrete valuation rings (More on Algebra, Definition more-algebra-definition-extension-discrete-valuation-rings, canonical tag 09E4). cited at statement lines 477-480. Defines the type of extension R'/R with induced fraction field and residue field extensions to which the lemma applies.
- Section on relative fields and base change maps for cycles (relative-cycles chapter, Section section-relative-fields, canonical tag 0H4E). cited at proof line 495. Makes sense of the vertical base change arrows via the identification X'_{K'} = X_K x_{Spec(K)} Spec(K') and similarly for closed fibres.
- Specialization maps on cycle modules over DVRs (relative-cycles chapter, Lemma lemma-specialization-module, canonical tag 0H4I). cited at proof lines 506-505. Computes sp_{X/R}([Z]) = [Zbar_kappa]_r and sp_{X'/R'}([Zbar_{K'}]_r) = [(Zbar_{R'})_{kappa'}]_r.

Source reference. Original definition 09E4: sources/stacks-chow/tags.txt, source lines 11887–11887. Cited at statement lines 477-480: type of the DVR extension R'/R.

Source reference. Original section 0H4E: sources/stacks-chow/tags.txt, source lines 21349–21349. Cited at proof line 495: the vertical arrows make sense as base change maps.

Source reference. Original lemma 0H4I: sources/stacks-chow/tags.txt, source lines 21353–21353. Cited at proof lines 501-505: computes both specialization maps.

Lemma 0.2. *Let R'/R be an extension of discrete valuation rings inducing fraction field extension K'/K and residue field extension κ'/κ (More on Algebra, Definition*

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09E4). Let X be locally of finite type over R. Denote $X' = X_{R'}$. Then the diagram

$$\begin{array}{ccc} Z_r(X'_{K'}) & \xrightarrow{sp_{X'/R'}} & Z_r(X'_{\kappa'}) \\ \uparrow & & \uparrow \\ Z_r(X_K) & \xrightarrow{sp_{X/R}} & Z_r(X_\kappa) \end{array}$$

commutes where $r \geq 0$ and the vertical arrows are base change maps.

Proof. Observe that $X'_{K'} = X_{K'} = X_K \times_{\mathrm{Spec}(K)} \mathrm{Spec}(K')$ and similarly for closed fibres, so that the vertical arrows indeed make sense (see Section 0H4E). Now if $Z \subset X_K$ is an integral closed subscheme with scheme theoretic image $\overline{Z} \subset X$, then we see that $Z_{K'} \subset X_{K'}$ is a closed subscheme with scheme theoretic image $\overline{Z}_{R'} \subset X_{R'}$. The base change of $[Z]$ is $[Z_{K'}]_r = [\overline{Z}_{K'}]_r$ by definition. We have

$$sp_{X/R}([Z]) = [\overline{Z}_\kappa]_r \quad \text{and} \quad sp_{X'/R'}([\overline{Z}_{K'}]_r) = [(\overline{Z}_{R'})_{\kappa'}]_r$$

by Lemma 0H4I. Since $(\overline{Z}_{R'})_{\kappa'} = (\overline{Z}_\kappa)_{\kappa'}$ we conclude. □